

Towards Higher Observational Type Theory

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What is the identity type?

- ▶ Per Martin-Löf:

$$\begin{aligned} \text{data Id}_{(A:\text{Type})} (a : A) : A \rightarrow \text{Type} \\ \text{refl} : \text{Id}_A a a \end{aligned}$$

- ▶ Cubical Type Theory: $\mathbb{I} \rightarrow A$
- ▶ Observational Type Theory (OTT):

$$\text{Id}_{A \times B} (a_0, b_0) (a_1, b_1) := \text{Id}_A a_0 a_1 \times \text{Id}_B b_0 b_1$$
$$\text{data Id}_{\mathbb{N}} : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \text{Prop}$$
$$\text{zero} : \text{Id}_{\mathbb{N}} \text{zero zero}$$
$$\text{suc} : \text{Id}_{\mathbb{N}} n_0 n_1 \rightarrow \text{Id}_{\mathbb{N}} (\text{suc } n_0) (\text{suc } n_1)$$
$$\text{codata Id}_{\text{Stream}_A} : \text{Stream}_A \rightarrow \text{Stream}_A \rightarrow \text{Prop}$$
$$\text{head} : \text{Id}_{\text{Stream}_A} s s' \rightarrow \text{Id}_A (\text{head } s) (\text{head } s')$$
$$\text{tail} : \text{Id}_{\text{Stream}_A} s s' \rightarrow \text{Id}_{\text{Stream}_A} (\text{tail } s) (\text{tail } s')$$
$$\text{Id}_{\text{Set}} A B := \text{equality of codes}$$

What is the identity type?

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- ▶ Cubical Type Theory: $\mathbb{I} \rightarrow A$
- ▶ **H**igher **O**bservational Type Theory (**HOTT**):

$$\text{Id}_{A \times B} (a_0, b_0) (a_1, b_1) := \text{Id}_A a_0 a_1 \times \text{Id}_B b_0 b_1$$
$$\text{data } \text{Id}_{\mathbb{N}} : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \text{Type}$$
$$\text{zero} : \text{Id}_{\mathbb{N}} \text{zero zero}$$
$$\text{suc} : \text{Id}_{\mathbb{N}} n_0 n_1 \rightarrow \text{Id}_{\mathbb{N}} (\text{suc } n_0) (\text{suc } n_1)$$
$$\text{codata } \text{Id}_{\text{Stream}_A} : \text{Stream}_A \rightarrow \text{Stream}_A \rightarrow \text{Type}$$
$$\text{head} : \text{Id}_{\text{Stream}_A} s s' \rightarrow \text{Id}_A (\text{head } s) (\text{head } s')$$
$$\text{tail} : \text{Id}_{\text{Stream}_A} s s' \rightarrow \text{Id}_{\text{Stream}_A} (\text{tail } s) (\text{tail } s')$$
$$\text{Id}_{\text{Type}} A B := A \simeq B$$

Promises of HOTT

- + Explainable
- + Definitional univalence:

$$\frac{\frac{f : A \rightarrow B \quad \text{isEquiv}(f)}{\text{univalence}(f) : \text{Id}_{\text{Type}} A B}}{\text{transport}_{\lambda X.X} (\text{univalence}(f)) = f : A \rightarrow B}$$

- + Efficient
- Some computation rules for higher inductive types will be typal

Why is OTT $\rightarrow$ HOTT difficult?

Computation in OTT:

$$\text{coe} : (A_0 A_1 : \text{Set}) \rightarrow \text{Id}_{\text{Set}} A_0 A_1 \rightarrow A_0 \rightarrow A_1$$

$$\text{coe } (A_0 \times B_0) (A_1 \times B_1) _ w := (\text{coe } A_0 A_1 _ (\text{fst } w), \text{coe } B_0 B_1 _ (\text{snd } w))$$

$$\text{ap}_f : \text{Id}_A a_0 a_1 \rightarrow \text{Id}_B (f a_0) (f a_1)$$

Why is OTT $\rightarrow$ HOTT difficult?

Computation in HOTT:

$$\text{coe} : (A_0 A_1 : \text{Type}) \rightarrow \text{Id}_{\text{Type}} A_0 A_1 \rightarrow A_0 \rightarrow A_1$$
$$\text{coe } A_0 A_1 (\text{univalence}(f, e)) a_0 := f(a_0)$$
$$\text{ap}_F : \text{Id}_{\text{Type}} A_0 A_1 \rightarrow \text{Id}_{\text{Type}} (F A_0) (F A_1)$$

Parametricity

- ▶ abstract types algebraically:
 - ▶ $\alpha : (X : \text{Type}) \rightarrow X \rightarrow X$ says that every type is equipped with a “set with endofunction” algebra
 - ▶ every function $f : A \rightarrow B$ is a “set with endofunction”-homomorphism, i.e. $f (\alpha_A a) = \alpha_B (f a)$
 - ▶ given an $a_0 : A$, we choose $f : A \rightarrow A$ with $f x := a_0$, then:

$$a_0 = f (\alpha_A a_0) = \alpha_A (f a_0) = \alpha_A a_0$$

- ▶ $\alpha : (N : \text{Type}) \rightarrow N \rightarrow (N \rightarrow N) \rightarrow N$ is not algebraic
- ▶ Reynolds (1983): preservation of relations:
 - ▶ $\alpha : (X : \text{Type}) \rightarrow X \rightarrow X$
 - ▶ every relation $R : A \rightarrow B \rightarrow \text{Type}$ is preserved: $R a b \rightarrow R (\alpha_A a) (\alpha_B b)$
 - ▶ if the relation is the graph of a function: $f a = b \rightarrow f (\alpha_A a) = \alpha_B b$

External to internal parametricity

- External parametricity: Reynolds (1983), Wadler (1989), Plotkin–Abadi (1993), Bernardy–Jansson–Paterson (2010)

$$\frac{A : \text{Type}}{A^P : A \rightarrow A \rightarrow \text{Type}} \qquad \frac{a : A}{a^P : A^P a a}$$

$$(A \rightarrow B)^P f_0 f_1 := (a_0 : A)(a_1 : A) \rightarrow A^P a_0 a_1 \rightarrow B^P (f_0 a_0) (f_1 a_1)$$

$$\text{Type}^P A_0 A_1 := A_0 \rightarrow A_1 \rightarrow \text{Type}$$

- Internal parametricity: Bernardy–Moulin (2012)

$$\frac{\frac{a_2 : A^P a_0 a_1}{a_2^P : (A^P a_0 a_1)^P a_2 a_2}}{a_2^P : A^{PP} a_0 a_0 a_0^P a_1 a_1 a_1^P a_2 a_2} \qquad \frac{\frac{(\lambda x.x^P) : (x : A) \rightarrow A^P x x}{(\lambda x.x^P)^P a_0 a_1 a_2 : (A^P x x)^P a_0^P a_1^P}}{(\lambda x.x^P)^P a_0 a_1 a_2 : A^{PP} a_0 a_1 a_2 a_0 a_1 a_2 a_0^P a_1^P}$$

$$\begin{array}{ccc} & a_1^P & \\ a_1 & \longrightarrow & a_1 \\ a_2 \uparrow & & \uparrow a_2 \\ & a_0^P & \\ a_0 & \longrightarrow & a_0 \end{array}$$

$$\begin{array}{ccc} & a_1 & \\ a_2 & \uparrow & \\ & a_0 & \end{array}$$

$$\begin{array}{ccc} & a_2 & \\ a_0 & \longrightarrow & a_1 \\ a_0^P \uparrow & & \uparrow a_1^P \\ & a_2 & \\ & a_0 & \longrightarrow & a_1 \end{array}$$

Theories of internal parametricity

- ▶ Bernardy–Moulin (2012, 2013): higher dimensional syntax
- ▶ Bernardy–Coquand–Moulin (2015): substructural interval
- ▶ Nuyts–Devriese (2018): simpler semantics
- ▶ Cavallo–Harper (2021): parametricity and univalence combined
- ▶ Altenkirch–Chamoun–Kaposi–Shulman (2024): first structural theory
- ▶ Mike Shulman (2024–2026): the proof assistant Narya
- ▶ Sarah Reboullet’s thesis (2025): analysis of a problem in BCM 2015
- ▶ Jem Lord (2025): internal axiomatic parametricity

Informal Narya

$$\frac{A : \text{Type}}{A^P : A \rightarrow A \rightarrow \text{Type}} \qquad \frac{a : A}{a^P : A^P a a}$$

$$\frac{R : A \rightarrow B \rightarrow \text{Type}}{\text{Gel } R : \text{Type}^P A B} \qquad \frac{A_2 : \text{Type}^P A_0 A_1}{A_2 : A_0 \rightarrow A_1 \rightarrow \text{Type}} \qquad \text{Gel } R a b \cong R a b$$

$$(A \rightarrow B)^P f_0 f_1 = \{a_0 a_1 : A\} \rightarrow A^P a_0 a_1 \rightarrow B^P (f_0 a_0) (f_1 a_1)$$

$$(\Sigma A B)^P (a_0, b_0) (a_1, b_1) = \Sigma (a_2 : A^P a_0 a_1). B^P a_2 b_0 b_1$$

$$(\lambda c. (a : A c) \rightarrow B (c, a))^P c_2 f_0 f_1 = (a_2 : A^P c_2 a_0 a_1) \rightarrow B^P (c_2, a_2) (f_0 a_0) (f_1 a_1)$$

$$x^P = x_2 \qquad (\lambda x. t)^P = \lambda \{x_0\} \{x_1\} x_2. t^P \qquad (t u)^P = t^P u^P$$

$$\frac{a_{22} : A^{PP} a_{02} a_{12} a_{20} a_{21}}{\text{sym } a_{22} : A^{PP} a_{20} a_{21} a_{02} a_{12}} \qquad \text{sym (sym } a_{22}) = a_{22} \qquad \dots$$

whiteboard example: polymorphic identity

Iteration → induction

data $\mathbb{N}$: Type

zero : $\mathbb{N}$

suc : $\mathbb{N} \rightarrow \mathbb{N}$

data $\mathbb{N}^P$: Type^P $\mathbb{N} \mathbb{N}$

zero^P : $\mathbb{N}^P$ zero zero

suc^P : $\mathbb{N}^P n_0 n_1$
 $\rightarrow \mathbb{N}^P (\text{suc } n_0) (\text{suc } n_1)$

ind _{$\mathbb{N}$} : (P : $\mathbb{N} \rightarrow \text{Type}$) $\rightarrow P$ zero $\rightarrow (P n \rightarrow P (\text{suc } n)) \rightarrow (n : \mathbb{N}) \rightarrow P n$

ind _{$\mathbb{N}$} $P z^P s^P n :=$

$$\begin{array}{c} \text{ite}_{\mathbb{N}^P} (\text{Glue } (n_0 n_1 \mapsto P n_0)) \quad \underbrace{z^P} \quad \underbrace{(n^P \mapsto s^P n^P)} \quad n^P \\ \underbrace{\text{Glue } (n_0 n_1 \mapsto P n_0) \text{ zero zero}}_{\cong P \text{ zero}} \quad \underbrace{\text{Glue } (n_0 n_1 \mapsto P n_0) n_0 n_1}_{\cong P n_0} \quad \underbrace{\text{Glue } (n_0 n_1 \mapsto P n_0) (\text{suc } n_0) (\text{suc } n_1)}_{\cong P (\text{suc } n_0)} \\ \hline : \text{Glue } (n_0 n_1 \mapsto P n_0) n n \cong P n \end{array}$$

ite _{$\mathbb{N}$} : (A : Type) $\rightarrow A \rightarrow (A \rightarrow A) \rightarrow \mathbb{N} \rightarrow A$

ite _{$\mathbb{N}$} $A z s$ zero = z

ite _{$\mathbb{N}$} $A z s$ (suc n) = s (ite _{$\mathbb{N}$} $A z s$ n)

ite _{$\mathbb{N}^P$} : (A^P : Type^P $\mathbb{N} \mathbb{N}$) $\rightarrow A^P$ zero zero
 $\rightarrow (A^P n_0 n_1 \rightarrow A^P (\text{suc } n_0) (\text{suc } n_1)) \rightarrow \mathbb{N}^P n_0 n_1$
 $\rightarrow A^P n_0 n_1$

ite _{$\mathbb{N}^P$} $A^P z^P s^P$ zero^P = z^P

ite _{$\mathbb{N}^P$} $A^P z^P s^P$ (suc^P n^P) = s^P (ite _{$\mathbb{N}^P$} $A^P z^P s^P$ n^P)

Semi-cubical types*

codata $T : \text{Type}$

$\text{ty} : T \rightarrow \text{Type}$

$\text{rel} : (x : T) \rightarrow T^P x x$

codata $T^P : \text{Type}^P T T$

$\text{ty}^P : (T \rightarrow \text{Type})^P \text{ty ty}$

$\text{rel}^P : ((x : T) \rightarrow T^P x x)^P \text{rel rel}$

codata $T^{PP} : (\text{Type}^P T T)^P T^P T^P$

$\text{ty}^{PP} : ((T \rightarrow \text{Type})^P \text{ty ty})^P \text{ty}^P \text{ty}^P$

$\text{rel}^{PP} : (((x : T) \rightarrow T^P x x)^P \text{rel rel})^P \text{rel}^P \text{rel}^P$

...

Semi-cubical types*

codata $T : \text{Type}$

$\text{ty} : T \rightarrow \text{Type}$

$\text{rel} : (x : T) \rightarrow T^P x x$

codata $T^P : \text{Type}^P T T$

$\text{ty}^P : T^P x_0 x_1 \rightarrow \text{Type}^P (\text{ty } x_0) (\text{ty } x_1)$

$\text{rel}^P : (x_2 : T^P x_0 x_1) \rightarrow T^{PP} x_2 x_2 (\text{rel } x_0) (\text{rel } x_1)$

codata $T^{PP} : \text{Type}^{PP} T^P T^P T^P T^P$

$\text{ty}^{PP} : T^{PP} x_{02} x_{12} x_{20} x_{21} \rightarrow \text{Type}^{PP} (\text{ty}^P x_{02}) (\text{ty}^P x_{12}) (\text{ty}^P x_{20}) (\text{ty}^P x_{21})$

$\text{rel}^{PP} : (x_{22} : T^{PP} x_{02} x_{12} x_{20} x_{21}) \rightarrow T^{PPP} x_{22} x_{22} (\text{rel}^P x_{02}) (\text{rel}^P x_{12}) (\text{rel}^P x_{20}) (\text{rel}^P x_{21})$

...

Fibrancy

codata isFib : Type → Type

coe : isFib^P A₂ f₀ f₁ → A₀ → A₁

coh : (f₂ : isFib^P A₂ f₀ f₁)(a₀ : A₀) → A₂ a₀ (coe f₂ a₀)

coe⁻¹ : isFib^P A₂ f₀ f₁ → A₁ → A₀

coh⁻¹ : (f₂ : isFib^P A₂ f₀ f₁)(a₁ : A₁) → A₂ (coe⁻¹ f₂ a₁) a₁

id : (f₂ : isFib^P A₂ f₀ f₁)(a₀ : A₀)(a₁ : A₁) → isFib (A₂ a₀ a₁)

Fib := Σ Type isFib

$$\frac{A : \text{Type} \quad f : \text{isFib } A}{\frac{f^P : \text{isFib}^P A^P f f}{\text{coe } f^P : A \rightarrow A}}$$

Two-dimensional Kan fillers

codata isFib : Type → Type

coe : isFib^P A₂ f₀ f₁ → A₀ → A₁ (also in the other direction)

coh : (f₂ : isFib^P A₂ f₀ f₁)(a₀ : A₀) → A₂ a₀ (coe f₂ a₀)

id : (f₂ : isFib^P A₂ f₀ f₁)(a₀ : A₀)(a₁ : A₁) → isFib (A₂ a₀ a₁)

id^P : (f₂₂ : isFib^{PP} A₂₂ f₀₂ f₁₂ f₂₀ f₂₁)(a₀₂ : A₀₂ a₀₀ a₀₁)(a₁₂ : A₁₂ a₁₀ a₁₁)
→ isFib^P (A₂₂ a₀₂ a₁₂) (id f₂₀ a₀₀ a₁₀) (id f₂₁ a₀₁ a₁₁)

$$\frac{\frac{f : \text{isFib } A}{f^{\text{PP}} : \text{isFib}^{\text{PP}} A^{\text{PP}} f^{\text{P}} f^{\text{P}} f^{\text{P}} f^{\text{P}}} \quad a_{02} : A^{\text{P}} a_{00} a_{01} \quad a_{12} : A^{\text{P}} a_{10} a_{11}}{\text{coe} (\text{id}^{\text{P}} f^{\text{PP}} a_{02} a_{12}) : A^{\text{P}} a_{00} a_{10} \rightarrow A^{\text{P}} a_{01} a_{11}}$$

$$\begin{array}{ccccc} & a_{01} & & a_{11} & \\ a_{02} & \uparrow & & \uparrow & a_{12} \\ & a_{00} & \xrightarrow{a_{20}} & a_{10} & \end{array}$$

Identity type

$$(A, f) : \text{Fib}$$

$$\text{Id}_{A,f} : A \rightarrow A \rightarrow \text{Fib}$$

$$\text{Id}_{A,f} a_0 a_1 := (A^P a_0 a_1, \text{id } f^P a_0 a_1)$$

$$B : A \rightarrow \text{Fib}$$

$$\text{snd} \circ B : (a : A) \rightarrow \text{isFib } (B a) \quad a_2 : A^P a_0 a_1$$

$$(\text{snd} \circ B)^P a_2 : \text{isFib}^P (B^P a^P) (\text{snd } (B a_0)) (\text{snd } (B a_1))$$

$$\text{transport}_B a_2 := \text{coe } ((\text{snd} \circ B)^P a_2) : \text{fst } (B a_0) \rightarrow \text{fst } (B a_1)$$

Assume $a : A$.

$$(\Sigma(x : A). A^P a x)^P (a, a^P) (b, e) = \Sigma(x_2 : A^P a b). A^{PP} a^P x_2 a^P e$$

$$\begin{array}{ccccc} a & \dashrightarrow & b \\ a^P \uparrow & \Uparrow & \uparrow e \\ a & \xrightarrow{a^P} & a \end{array}$$

Fib closed under $\top$, $\perp$, $\times$, Σ , Π , Bool

$$\begin{array}{l} F : \text{Type} \rightarrow \text{Type} \\ \text{coe} : F^P C_2 f_0 f_1 \rightarrow C_0 \rightarrow C_1 \quad (\text{also in other dir.}) \\ \text{coh} : (f_2 : F^P C_2 f_0 f_1)(c_0 : C_0) \rightarrow C_2 c_0 (\text{coe } f_2 c_0) \\ \text{id} : F^P C_2 f_0 f_1 \rightarrow (c_0 : C_0)(c_1 : C_1) \rightarrow F (C_2 c_0 c_1) \\ \hline \text{gen}_{\text{isFib}} : (A : \text{Type}) \rightarrow F A \rightarrow \text{isFib } A \end{array}$$

whiteboard

Lemma: $\cong^P A_2 B_2 e_0 e_1 \rightarrow (a_0 : A_0)(a_1 : A_1) \rightarrow A_2 a_0 a_1 \cong B_2 (e_0 a_0) (e_1 a_1)$.

Fib closed under W and M

$$\begin{array}{lcl} S & : & \text{Type} \\ P & : & S \rightarrow \text{Type} \\ \hline W & : & \text{Type} \\ \text{sup} & : & (s : S) \rightarrow (P\ s \rightarrow W) \rightarrow W \end{array}$$

Indexed W-types:

$$\begin{array}{lcl} I & : & \text{Type} \\ S & : & \text{Type} \\ P & : & S \rightarrow \text{Type} \\ j & : & \{s : S\} \rightarrow P\ s \rightarrow I \\ k & : & S \rightarrow I \\ \hline W & : & I \rightarrow \text{Type} \\ \text{sup} & : & (s : S) \rightarrow ((p : P\ s) \rightarrow W\ (j\ p)) \\ & & \rightarrow W\ (k\ s) \end{array}$$

Non-uniform parameterised W-types:

$$\begin{array}{lcl} I & : & \text{Type} \\ S & : & I \rightarrow \text{Type} \\ P & : & \{i : I\} \rightarrow S\ i \rightarrow \text{Type} \\ \\ k & : & \{i : I\} \{s : S\ i\} \rightarrow P\ i\ s \rightarrow I \\ \hline W & (i : I) & : \text{Type} \\ \text{sup} & : & (s : S\ i) \rightarrow ((p : P\ s) \rightarrow W\ (k\ p)) \\ & & \rightarrow W\ i \end{array}$$

Univalence

$$\text{is11} (R : A \rightarrow B \rightarrow \text{Fib}) := ((a : A) \rightarrow \text{isContr} (\Sigma(b : B).R a b)) \times \\ ((b : B) \rightarrow \text{isContr} (\Sigma(a : A).R a b))$$

$\text{codata isBisim} (R : A \rightarrow B \rightarrow \text{Fib}) : \text{Type}$

$\text{coe} : \text{isBisim } R \rightarrow A \rightarrow B$

(also in the other direction)

$\text{coh} : (x : \text{isBisim } R)(a : A) \rightarrow R a (\text{coe } x a)$

$\text{id} : \text{isBisim}^R R_2 x_0 x_1 \rightarrow (r_0 : R_0 a_0 b_0)(r_1 : R_1 a_1 b_1) \rightarrow \text{isBisim} (a_2 b_2 \mapsto R_2 a_2 b_2 r_0 r_1)$

$$\frac{\begin{array}{ccc} B_0 & \xrightarrow{B_2} & B_1 \\ R_0, \text{is11} \uparrow & R_2, \text{is11}^P \uparrow & R_1, \text{is11} \uparrow \\ A_0 & \xrightarrow{A_2} & A_1 \end{array}}{\text{is11} (a_2 b_2 \mapsto R_2 a_2 b_2 r_0 r_1)}$$

$\text{is11} \xrightarrow{\quad} \text{isBisim} \xrightleftharpoons[\text{R} := \text{id}]{\text{Gel } R} \text{Fib}^P A_0 A_1$

Summary

- ▶ Narya: a proof assistant with internal parametricity
 - ▶ polymorphic identity
 - ▶ iterator $\rightarrow$ induction, with definitional computation rules
 - ▶ symmetric inductive and coinductive types
 - ▶ semi-cubical types*
 - ▶ higher coinductive types, fibrancy
 - ▶ fibrant families come with transport
 - ▶ $\top$, $\perp$, $\times$, Σ , Π , Bool, W, M fibrant
 - ▶ definitional univalence
- ▶ Future:
 - ▶ isFib Fib,
 - ▶ normalisation
 - ▶ higher inductive types

Uniformity

